

Question Paper

Exam Date & Time: 27-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Software Engineering [IT_4061]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the evolution of software development from an art form to an engineering discipline. How did this transformation impact software quality and maintainability? (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

B) Apply the concepts of software process models to demonstrate how the Waterfall, Spiral, and Agile models differ in terms of flexibility, risk handling, and customer involvement. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Examine how SSDLC extends the traditional SDLC and analyze its advantages in modern software development environments. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

B) Analyze how incomplete or ambiguous Software Requirement Specifications (SRS) can lead to project failures. Illustrate with an example. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply the concept of layered design architecture to demonstrate how it enhances system modularity and fault isolation in large-scale applications. (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Application"

B) Construct a DFD model for a payroll management system that clearly shows the identification of data transformations and process divisions. (5)

Course Outcome: "CO3" / Cognitive Level: "Create"
- 4) Develop a software design model that applies structured design principles to show how complexity can be reduced and maintainability enhanced. (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Create"

B) Analyze how different UML diagrams (use case, class, sequence) complement each other in representing a complete system design. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Evaluate the effectiveness of state-chart diagrams in modeling dynamic system behavior, using an ATM transaction as an example. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Assess the effectiveness of code review in complementing automated testing to improve software reliability and maintainability. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4061:Software Engineering

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	5
3	3	CO 3	15
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4061:Software Engineering

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(20%)
4	Analysis	4	20(40%)
5	Create	2	10(20%)
6	Evaluate	2	10(20%)
Total		10	50 (100%)